permanent team of full-time researchers, which became tightly knit and extremely powerful, and which might even stay with him until the end of his career.

Some of my Soviet friends attributed this difference to the failings of the British/American system: Almost all great British or American physicists work at universities, where research is often subservient to teaching and where there are inadequate numbers of permanent positions available to permit building up a strong, lasting group of researchers. As a result, there have been *no* theoretical physics research groups in Britain or America that can pretend to be the equal of Landau's group in the 1930s through 1950s, or of Zel'dovich's group in the 1960s and 1970s. The West, in this sense, had no hope of competing with the Soviet Union.

Some of my American friends attributed the difference to the failings of the Soviet system: It was very difficult, logistically, to move from institute to institute and city to city in the U.S.S.R., so young physicists were forced to remain with their mentors; they had no opportunity to get out and start independent groups of their own. The result, the critics asserted, was a feudal system. The mentor was like a lord and his team like serfs, indentured for most of their careers. The lord and serfs were interdependent in a complex way, but there was no question who was boss. If the lord was a master craftsman like Zel'dovich or Landau, the lord/serf team could be richly productive. If the lord was authoritarian and not so outstanding (as was commonly the case), the result could be tragic: a waste of human talent and a miserable life for the serfs.

In the Soviet system, each great mentor such as Zel'dovich produced just one research team, albeit a tremendously powerful one, one unequaled anywhere in the West. By contrast, great American or British mentors like Wheeler and Sciama produce as their progeny many smaller and weaker research groups, scattered throughout the land, but those groups can have a large cumulative impact on physics. The American and British mentors have a constant influx of new, young people to help keep their minds and ideas fresh. In those rare cases where Soviet mentors wanted to start over afresh, they had to break their ties with their old team in a manner which could be highly traumatic.

This, in fact, was destined to happen to Zel'dovich: He began building his astrophysics team in 1961; by 1964 it was superior to any other theoretical astrophysics team anywhere in the world; then in 1978, soon